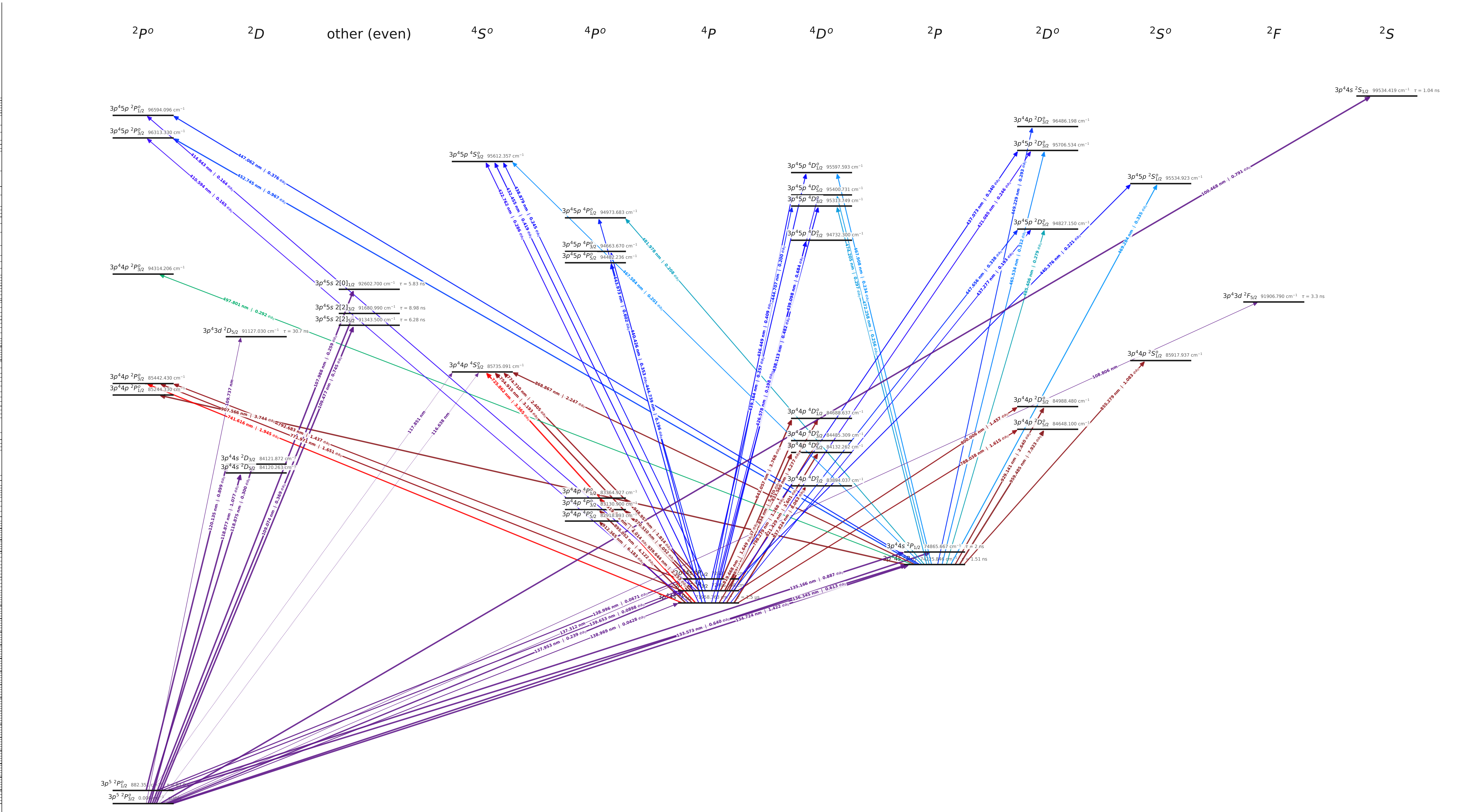


Chlorine

Cl (Cl I). Strongest classified transitions below 2 μm (NIST): 43 levels, 79 lines.

Cl⁺ ionisation limit 104591.01 cm⁻¹ = 12.96763 eV (95.611 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[U erU'] (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3p ⁵ ²P _{3/2} - 3p ⁴ 4s ²S _{1/2}	100.4678	-	2983.9668	0.791	6.250e+08	153	65	interstellar Cl I 1005 Å absorption line
3p ⁵ ²P _{3/2} - 3p ⁴ 5s 2[0] _{1/2}	107.9882	-	2776.1591	0.259	5.380e+07	27.3	31.4	interstellar Cl I 1080 Å absorption line
3p ⁵ ²P _{3/2} - 3p ⁴ 3d ²F _{5/2}	108.8059	-	2755.2962	-	-	48.23	-	interstellar Cl I 1088 Å line
3p ⁵ ²P _{3/2} - 3p ⁴ 5s 2[2] _{3/2}	109.0739	-	2748.5269	0.369	5.330e+07	17.72	47.9	interstellar Cl I 1091 Å absorption line
3p ⁵ ²P _{3/2} - 3p ⁴ 5s 2[2] _{1/2}	109.4769	-	2738.4092	0.745	1.430e+08	25.34	89.8	interstellar Cl I 1095 Å absorption line
3p ⁵ ²P _{3/2} - 3p ⁴ 3d ²D _{3/2}	109.7369	-	2731.9196	-	-	5.184	-	interstellar Cl I 1097 Å line
3p ⁵ ²P _{3/2} - 3p ⁴ 4p ⁴S _{3/2}	116.6384	-	2570.2734	-	-	-	-	two-photon LIF
3p ⁵ ²P _{1/2} - 3p ⁴ 4p ⁴S _{3/2}	117.8512	-	2543.8211	-	-	-	-	TALIF detection of spin-orbit excited Cl
3p ⁵ ²P _{3/2} - 3p ⁴ 4s ²D _{3/2}	118.8751	-	2521.9103	0.300	2.710e+07	-	-	VUV resonance line of the 3p5 - 3p4 4s array
3p ⁵ ²P _{3/2} - 3p ⁴ 4s ²D _{5/2}	118.8774	-	2521.8620	1.077	2.330e+08	-	-	VUV resonance line of the 3p5 - 3p4 4s array
3p ⁵ ²P _{1/2} - 3p ⁴ 4s ²D _{3/2}	120.1352	-	2495.4581	0.899	2.360e+08	-	-	VUV resonance line of the 3p5 - 3p4 4s array
3p ⁵ ²P _{3/2} - 3p ⁴ 4s ²P _{1/2}	133.5726	-	2244.4162	0.640	1.740e+08	79.58	34.8	VUV resonance line of the 3p5 - 3p4 4s array
3p ⁵ ²P _{3/2} - 3p ⁴ 4s ²P _{3/2}	134.7240	-	2225.2349	1.422	4.190e+08	105.4	85.2	134.72 nm VUV resonance line
3p ⁵ ²P _{1/2} - 3p ⁴ 4s ²P _{1/2}	135.1656	-	2217.9640	0.887	3.230e+08	79.58	64.6	VUV resonance line of the 3p5 - 3p4 4s array
3p ⁵ ²P _{1/2} - 3p ⁴ 4s ²P _{3/2}	136.3447	-	2198.7827	0.613	7.500e+07	105.4	14.8	136.34 nm VUV line from the metastable 2P1/2 level
3p ⁵ ²P _{3/2} - 3p ⁴ 4s ⁴P _{1/2}	137.3116	-	2183.2997	-	-	-	-	VUV resonance line of the 3p5 - 3p4 4s array
3p ⁵ ²P _{3/2} - 3p ⁴ 4s ⁴P _{3/2}	137.9528	-	2173.1526	0.239	1.100e+07	-	-	VUV resonance line of the 3p5 - 3p4 4s array
3p ⁵ ²P _{3/2} - 3p ⁴ 4s ⁴P _{5/2}	138.9693	-	2157.2575	0.043	2.300e+05	0.1061	34.5	VUV resonance line of the 3p5 - 3p4 4s array
3p ⁴ 4s ⁴P _{3/2} - 3p ⁴ 4p ⁴S _{3/2}	725.8618	725.6618	413.0159	3.365	1.500e+07	-	-	725.9 nm

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).
Arrow colour = vacuum wavelength (nm)
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.

³⁵Cl hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
3p ⁵ ²P _{3/2}	205.04687(3)	54.87290(6)	F=0: -700.34 F=1: -550.16 F=2: -194.94 F=3: +475.07
3p ⁵ ²P _{1/2}	1037.209(14)	-	F=1: -1296.51 F=2: +777.91

Measured A, B constants; sources in the data file.